Lecture 7

January 16, 2026

Moments

- The study of probability distributions of a random variable is essentially the study of some numerical characteristics/parameters associated with them.
- These distribution parameters play a key role in mathematical statistics.
- A comprehensive study of moments allow us to investigate different characteristics of these parameters.
- One can deal with different measures of central tendency, measures of dispersion, shapes of distribution, etc. using the concept of moments.

Expected Value of a Discrete Random Variable

Let X be a random variable of the discrete type with probability mass function (PMF), $p_i = P(X = x_i)$, $i = 1, 2, \ldots$. If

$$\sum_{i=1}^{\infty} |x_i| p_i < \infty, \tag{1}$$

we say that the expected value (or mean), $E(X)$ of X exists and write

$$\mu = E(X) = \sum_{i=1}^{\infty} x_i p_i. \tag{2}$$

Note that the series $\sum_{i=1}^{\infty} x_i p_i$ may converge but the series $\sum_{i=1}^{\infty} |x_i| p_i$ may not. In that case we say that $E(X)$ does not exist.

Example 1

Let the PMF of a random variable X be given by

$$p_i = P\left\{X = (-1)^{i+1}\frac{3^i}{i}\right\} = \frac{2}{3^i}, \quad i = 1, 2, \ldots$$

Then

$$\sum_{i=1}^{\infty} |x_i| p_i = \sum_{i=1}^{\infty} \frac{2}{i} = \infty,$$

and $E(X)$ does not exist, although the series

$$\sum_{i=1}^{\infty} x_i p_i = \sum_{i=1}^{\infty} (-1)^{i+1} \frac{2}{i}$$

is convergent.

Expected Value of a Continuous Random Variable

Let X be a continuous random variable with probability density function (PDF), f. We say that the expected value (or mean), $E(X)$ of X exists, and is given by

$$\mu = E(X) = \int x f(x)\, dx, \tag{3}$$

provided that

$$\int |x| f(x)\, dx < \infty. \tag{4}$$

Important Remark

A similar definition can be given for the mean of any Borel-measurable function $h(X)$ of X. Thus if X is a continuous random variable having PDF f, we say that $E[h(X)]$ exists and is given by

$$\mu = E[h(X)] = \int h(x)f(x)\,dx, \tag{5}$$

provided that

$$\int |h(x)|f(x)\,dx < \infty. \tag{6}$$

Example 2

Let us consider the PDF

$$f(x) = \frac{1}{\pi(1+x^2)}, \quad -\infty < x < \infty.$$

Clearly,

$$\int_{-\infty}^{\infty} x f(x)\, dx = \int_{-\infty}^{\infty} \frac{x}{\pi(1+x^2)}\, dx = 0.$$

However, $E(X)$ does not exist since the integral

$$\int_{-\infty}^{\infty} \frac{|x|}{\pi(1+x^2)}\, dx$$

diverges.